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Skill 24.1 Exercise 1

The movies dataframe below is composed of the television shows and movies hosted on the Netflix platform in 2019

	type	title	country	release_year	rating	duration
0	Movie	Norm of the ...	United States	missing	PG	91.071
1	Movie	Jandino: Wha...	United Kingdom	2016	R	94.516
2	TV Show	Transformers...	United States	2013	G	1.127
3	TV Show	Transformers...	United States	2016	TV-14	1.68699999999999
4	Movie	#realityhigh...	United States	2017	TV-14	99.248

For each variable in the *movies* DataFrame, indicate whether it is *numerical* or *categorical*

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Skill 24.2 Exercise 1

The line of code below prints the different countries associated with the *country* variable from the *movies* DataFrame above.

Code
<code>print(movies.country.value_counts())</code>
Output
United States 2818 India 972 United Kingdom 419 Japan 245 South Korea 199 ... Romania, United Kingdom 1 Australia, United Kingdom, United Arab Emirates, Canada 1 Germany, United States, Hong Kong, Singapore 1 Mexico, Brazil 1 United Kingdom, Jordan, Qatar, Iran 1 Name: country, Length: 748, dtype: int64

For each categorical variable you identified in the previous exercise, indicate whether it is ordinal, nominal, or binary

Categorical Variable	Ordinal, nominal, or binary?
Type	

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Title	
Country	
Rating	

Skill 24.3 Exercise 1

Below is revised version of the movies DataFrame,

	show_id	type	title	country	release_year	rating	duration	cast_count	est_budget (USD)
0	s1	Movie	Dick Johnson Is Dead	United States	2020	PG-13	90 min	nan	11959075
1	s2	TV Show	Blood & Water	South Africa	2021	TV-MA	2 Seasons	19.0	9436627
2	s3	TV Show	Ganglands	nan	2021	TV-MA	1 Season	9.0	5719277
3	s4	TV Show	Jailbirds New Orleans	nan	2021	TV-MA	1 Season	nan	7881417
4	s5	TV Show	Kota Factory	India	2021	TV-MA	2 Seasons	8.0	27387486

For each numerical variable, indicate whether it is discrete or continuous

release_year	
cast_count	
est_budget (USD)	

Skill 24.4 Exercise 1

Below is a list of data types for the variables in the movies Data Frame

show_id	object
type	object
title	object
country	object
release_year	float64
rating	object
duration	object
cast_count	float64
est_budget (USD)	int64

For each variable below, use the *astype()* method to convert it to data type indicated.

release_year	int64	
cast_count	int64	
est_budget (USD)	float64	

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Skill 24.5 Exercise 1

Refer to the movies DataFrame

	show_id	type	title	country	release_year	rating	duration	cast_count	est_budget (USD)
0	s1	Movie	Dick Johnson Is Dead	United States	2020	PG-13	90 min	nan	11959075
1	s2	TV Show	Blood & Water	South Africa	2021	TV-MA	2 Seasons	19.0	9436627
2	s3	TV Show	Ganglands	nan	2021	TV-MA	1 Season	9.0	5719277
3	s4	TV Show	Jailbirds New Orleans	nan	2021	TV-MA	1 Season	nan	7881417
4	s5	TV Show	Kota Factory	India	2021	TV-MA	2 Seasons	8.0	27387486

Below are the categorical variables from a previous exercise. Choose the most appropriate *storage data type* for each variable from the following options: binary, category, or string. Justify each choice.

Categorical Variable	Current data type	Storage data type	Justification
Type	object		
Title	object		
Country	object		
Rating	object		

Use the `astype()` method to convert each variable above.

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Skill 24.6 Exercise 1

Refer to the movies DataFrame

	show_id	type	title	country	release_year	rating	duration	cast_count	est_budget (USD)
0	s1	Movie	Dick Johnson Is Dead	United States	2020	PG-13	90 min	nan	11959075
1	s2	TV Show	Blood & Water	South Africa	2021	TV-MA	2 Seasons	19.0	9436627
2	s3	TV Show	Ganglands	nan	2021	TV-MA	1 Season	9.0	5719277
3	s4	TV Show	Jailbirds New Orleans	nan	2021	TV-MA	1 Season	nan	7881417
4	s5	TV Show	Kota Factory	India	2021	TV-MA	2 Seasons	8.0	27387486

In a previous example, we converted the rating variable to a category data type. In the space below,

- Create a ratings list with the following values: 'NR', 'G', 'PG', 'PG-13', 'R'
- Use the `cat` method to access the categories
- Use the `set_categories` method to set the order of the categories

Skill 24.7 Exercise 1

Refer to the movies DataFrame

	show_id	type	title	country	release_year	rating	duration	cast_count	est_budget (USD)
0	s1	Movie	Dick Johnson Is Dead	United States	2020	PG-13	90 min	nan	11959075
1	s2	TV Show	Blood & Water	South Africa	2021	TV-MA	2 Seasons	19.0	9436627
2	s3	TV Show	Ganglands	nan	2021	TV-MA	1 Season	9.0	5719277
3	s4	TV Show	Jailbirds New Orleans	nan	2021	TV-MA	1 Season	nan	7881417
4	s5	TV Show	Kota Factory	India	2021	TV-MA	2 Seasons	8.0	27387486

In a previous example, we converted the rating variable to a category data type and set the order. Unfortunately, this isn't the best way to organize this data because it implies a relationship that doesn't necessarily exist.

In the space below, One-Hot Encode the ratings variable with `pd.get_dummies()`

What is the purpose of OHE?